THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET B

Chapter 3: **Metals and Non-metals** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The property of metals by which they can be beaten into thin sheets is: 1

(a) lustre **(b)** ductility **(c)** sonority **(d)** malleability

2. Which metal is the poorest conductor of heat? 1

(a) copper **(b)** aluminium **(c)** silver **(d)** lead

3. Which of the following reacts with cold water? 1

(a) iron **(b)** sodium **(c)** aluminium **(d)** zinc

4. A student states: “Non-metals never conduct electricity.” The correct response is: 1

(a) Correct
(b) Incorrect — graphite conducts electricity
(c) Correct for gases
(d) Cannot be determined

5. An element forms an oxide that turns blue litmus red. Student A says it is a metal; Student B says it is a non-metal. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. The ore of mercury is: 1

(a) bauxite **(b)** calamine **(c)** haematite **(d)** cinnabar (HgS)

7. Which metal is used to galvanise iron? 1

(a) copper **(b)** aluminium **(c)** tin **(d)** zinc

8. Which of these displaces copper from copper sulphate solution? 1

(a) gold **(b)** silver **(c)** mercury **(d)** zinc

9. The alloy of copper and tin is: 1

(a) brass **(b)** steel **(c)** bronze **(d)** solder

10. An ionic bond is formed by: 1

(a) transfer of electrons **(b)** sharing of electrons
(c) transfer of neutrons **(d)** sharing of protons

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why does calcium float when it reacts with water? 2

12. Name two metals that are found in the free state in nature. 2

13. What is the difference between an ore and a mineral? 2

14. A student says pure gold (24 carat) is ideal for making jewellery. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Write the electron-dot structures of sodium, oxygen and magnesium, and show the formation of MgO. 3

16. What happens when 3
(i) sodium is dropped in water
(ii) iron filings are added to copper sulphate solution
(iii) copper is heated in air? Write equations.

17. What is corrosion? How can it be prevented? Why is galvanised iron preferred over tinned iron for buckets? 3

OR

Explain the following: 3
(i) Metals like Na and K are extracted by electrolysis.
(ii) Hydrogen is not evolved when copper reacts with dilute HCl.
(iii) Carbon cannot reduce oxides of sodium.

18. What is roasting and calcination? Give one example of each with equations. 3

OR

Give one use each of 3
(i) aluminium
(ii) copper
(iii) iron, stating the property responsible.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Explain how mercury is obtained from cinnabar and copper from copper glance, with equations. 5
(ii) Why are these metals extracted by heating alone?

OR

(i) What are alloys? 5
(ii) Why are alloys made?
(iii) Name the constituents of brass, bronze and solder.
(iv) Give one use of solder.

20. **Case study:** A student added pieces of metals A, B, C and D to solutions as shown: A to $FeSO_4$ — no change; B to $CuSO_4$ — brown deposit; C to $ZnSO_4$ — no change; D to $FeSO_4$ — green colour fades. 5
(i) Which is the most reactive?
(ii) Which is the least reactive?
(iii) Arrange A, B and D in decreasing order of reactivity if B is iron.
(iv) What is the brown deposit?

OR

(i) What are ionic compounds? 5
(ii) Give four properties of ionic compounds.
(iii) Why are they called electrovalent compounds?

END OF QUESTION PAPER — **Total: 40 Marks**

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